

Loan Analysis

Import libraries

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

Load dataset

```
loan_data = pd.read_excel('loandataset.xlsx')
customer_data = pd.read_csv('customer_data.csv', sep=';')
```

Display the first few rows of our dataset

```
loan_data.head()
customer_data.head()
```

	id	first_name	last_name	email	gender	\
0	C1	Rafaelia	Constable	rconstable0@google.co.jp	Female	
1	C2	Rayner	Langhor	rlanghor1@arstechnica.com	Male	
2	C3	Felike	Duchesne	fduchesne2@blog.com	Male	
3	C4	Kelli	Winchcomb	kwinchcomb3@tuttocitta.it	Female	
4	C5	Timi	Butterley	tbutterley4@jalbum.net	Female	

		city	country
0		Kazanskoye	Russia
1		Yauca	Peru
2	Monte Santo de Minas		Brazil
3	Červené Pečky	Czech Republic	
4	Lorino		Russia

Merging two data frames

```
complete_data = pd.merge(loan_data, customer_data,
left_on='customerid', right_on='id')
complete_data.head()
```

	loan id	customerid	credit.policy		purpose	int.rate	\
0	193	C193	1	debt_consolidation	0.0743		
1	114	C114	1	debt_consolidation	0.0712		
2	920	C920	1	all_other	0.0800		
3	56	C56	1	small_business	0.1059		
4	93	C93	1	debt_consolidation	0.0870		

	installment	log.annual.inc	dti	fico	days.with.cr.line	...	\
0	178.68	11.461632	1.20	812	16259.04167	...	
1	64.96	11.608236	0.22	812	16213.00000	...	

2	313.37	10.274672	2.24	792	14100.00000	...
3	398.69	11.002100	17.12	747	14008.95833	...
4	383.87	11.918391	4.80	767	13349.95833	...

	delinq.2yrs	pub.rec	not.fully.paid	id	first_name	last_name \
0	0	0	0	C193	Yuma	Malecky
1	0	0	0	C114	Steven	Thalmann
2	0	0	0	C920	Yetta	Cupitt
3	1	0	0	C56	Mycah	Drohun
4	0	0	0	C93	Anita	Harcus

	email	gender	city	country
0	ymalecky5c@constantcontact.com	Male	Lavradio	Portugal
1	sthalmann35@bloglines.com	Male	Rio Meão	Portugal
2	ycupittpj@etsy.com	Female	Muruni	Indonesia
3	mdrohun1j@paypal.com	Male	Tendō	Japan
4	aharcus2k@jiathis.com	Female	NaN	NaN

[5 rows x 23 columns]

Cleaning Data in Python

```
# Check for missing data
complete_data.isnull().sum()

# Remove the rows with missing data
complete_data = complete_data.dropna()
complete_data.isnull().sum()

# Check for duplicated data
complete_data.duplicated().sum()

# Drop duplicated rows
complete_data = complete_data.drop_duplicates()

complete_data.head()
```

	loan id	customerid	credit.policy	purpose	int.rate \
0	193	C193	1	debt_consolidation	0.0743
1	114	C114	1	debt_consolidation	0.0712
2	920	C920	1	all_other	0.0800
3	56	C56	1	small_business	0.1059
5	503	C503	1	credit_card	0.0832

	installment	log.annual.inc	dti	fico	days.with.cr.line	...	\
0	178.68	11.461632	1.20	812	16259.04167	...	
1	64.96	11.608236	0.22	812	16213.00000	...	
2	313.37	10.274672	2.24	792	14100.00000	...	
3	398.69	11.002100	17.12	747	14008.95833	...	
5	160.57	10.985293	7.08	767	13319.04167	...	

	delinq.2yrs	pub.rec	not.fully.paid	id	first_name	last_name	\
0	0	0	0	C193	Yuma	Malecky	
1	0	0	0	C114	Steven	Thalmann	
2	0	0	0	C920	Yetta	Cupitt	
3	1	0	0	C56	Mycah	Drohun	
5	0	0	0	C503	Dorena	Keely	

	email	gender	city	country
0	ymalecky5c@constantcontact.com	Male	Lavrado	Portugal
1	sthalmann35@bloglines.com	Male	Rio Meão	Portugal
2	ycupittpj@etsy.com	Female	Muruni	Indonesia
3	mdrohunlj@paypal.com	Male	Tendō	Japan
5	dkeelydy@alibaba.com	Female	Oued Lill	Tunisia

[5 rows x 23 columns]

Functions in Python

```
# Define a function to categorize purpose into categories
# There are many purposes in the purpose column.
# this function takes the purpose column as input and output the
broad categories
```

```
def categorize_purpose(purpose):
    if purpose in ['credit_card', 'debt_consolidation']:
        return 'Financial'
    elif purpose in ['educational', 'small_business']:
        return 'Educational/Business'
    else:
        return 'Other'
```

```
categorize_purpose('credit_card')
```

```
complete_data['purpose_category'] =
complete_data['purpose'].apply(categorize_purpose)
```

```
complete_data.head()
```

	loan id	customerid	credit.policy		purpose	int.rate	\
0	193	C193	1	debt_consolidation	0.0743		
1	114	C114	1	debt_consolidation	0.0712		
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	installment	log.annual.inc	dti	fico	days.with.cr.line	...
0	178.68	11.461632	1.20	812	16259.04167	...
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2	313.37	10.274672	2.24	792	14100.00000	...
3	398.69	11.002100	17.12	747	14008.95833	...
5	160.57	10.985293	7.08	767	13319.04167	...

	not.fully.paid	id	first_name	last_name	\
0	0	C193	Yuma	Malecky	
1	0	C114	Steven	Thalmann	
2	0	C920	Yetta	Cupitt	
3	0	C56	Mycah	Drohun	
5	0	C503	Dorena	Keely	

	email	gender	city	country	\
0	ymalecky5c@constantcontact.com	Male	Lavrado	Portugal	
1	sthalmann35@bloglines.com	Male	Rio Meão	Portugal	
2	ycupittpj@etsy.com	Female	Muruni	Indonesia	
3	mdrohunlj@paypal.com	Male	Tendō	Japan	
5	dkeelydy@alibaba.com	Female	Oued Lill	Tunisia	

	purpose_category
0	Financial
1	Financial
2	Other
3	Educational/Business
5	Financial

[5 rows x 24 columns]

Creating a conditional statement function

```
def check_number(number):
    if number < 0:
        return 'Negatif'
    if number > 0:
        return 'Positif'
    else:
```

```

        return 'Zero'

result = check_number(5)

# Create a new function based on criteria
# dti ratio > 20 and the delinq.2years > 2 and the revol.util > 60
# then the borrower is on high risk

def assess_risk(row):
    if row['dti'] > 20 and row['delinq.2yrs'] > 2 and
row['revol.util'] > 60:
        return 'High Risk'
    else:
        return 'Low Risk'

complete_data['Risk Level'] = complete_data.apply(assess_risk, axis=1)

complete_data.head()

```

	loan id	customerid	credit.policy		purpose	int.rate	\
0	193	C193	1	debt_consolidation	0.0743		
1	114	C114	1	debt_consolidation	0.0712		
2	920	C920	1	all_other	0.0800		
3	56	C56	1	small_business	0.1059		
5	503	C503	1	credit_card	0.0832		

	installment	log.annual.inc	dti	fico	days.with.cr.line	...	\
0	178.68	11.461632	1.20	812	16259.04167	...	
1	64.96	11.608236	0.22	812	16213.00000	...	
2	313.37	10.274672	2.24	792	14100.00000	...	
3	398.69	11.002100	17.12	747	14008.95833	...	
5	160.57	10.985293	7.08	767	13319.04167	...	

	not.fully.paid	id	first_name	last_name	\
0	0	C193	Yuma	Malecky	
1	0	C114	Steven	Thalmann	
2	0	C920	Yetta	Cupitt	
3	0	C56	Mycah	Drohun	
5	0	C503	Dorena	Keely	

	email	gender	city	country	\
0	ymalecky5c@constantcontact.com	Male	Lavrado	Portugal	
1	sthalmann35@bloglines.com	Male	Rio Meão	Portugal	
2	ycupittpj@etsy.com	Female	Muruni	Indonesia	
3	mdrohunlj@paypal.com	Male	Tendō	Japan	
5	dkeelydy@alibaba.com	Female	Oued Lill	Tunisia	

	purpose_category	Risk Level
0	Financial	Low Risk
1	Financial	Low Risk

```

2         Other      Low Risk
3 Educational/Business Low Risk
5         Financial  Low Risk

```

```
[5 rows x 25 columns]
```

```
# Create a function to categorize the FICO score based on different ranges
```

```

def categorize_fico(fico_score):
    if fico_score >= 800 and fico_score <= 850:
        return 'Excellent'
    elif fico_score >= 740 and fico_score < 800:
        return 'Very Good'
    elif fico_score >= 670 and fico_score < 740:
        return 'Good'
    elif fico_score >= 580 and fico_score < 670:
        return 'Fair'
    else:
        return 'Poor'

```

```

complete_data['FICO Category'] =
complete_data['fico'].apply(categorize_fico)

```

```
complete_data.head()
```

	loan id	customerid	credit.policy		purpose	int.rate	\
0	193	C193	1	debt_consolidation	0.0743		
1	114	C114	1	debt_consolidation	0.0712		
2	920	C920	1	all_other	0.0800		
3	56	C56	1	small_business	0.1059		
5	503	C503	1	credit_card	0.0832		

	installment	log.annual.inc	dti	fico	days.with.cr.line	...
id \						
0	178.68	11.461632	1.20	812	16259.04167	...
C193						
1	64.96	11.608236	0.22	812	16213.00000	...
C114						
2	313.37	10.274672	2.24	792	14100.00000	...
C920						
3	398.69	11.002100	17.12	747	14008.95833	...
C56						
5	160.57	10.985293	7.08	767	13319.04167	...
C503						

	first_name	last_name		email	gender	
city \						
0	Yuma	Malecky	ymalecky5c@constantcontact.com		Male	
Lavradio						
1	Steven	Thalmann	sthalmann35@bloglines.com		Male	Rio

Meão					
2	Yetta	Cupitt	ycupittpj@etsy.com	Female	
Muruni					
3	Mycah	Drohun	mdrohunlj@paypal.com	Male	
Tendō					
5	Dorena	Keely	dkeelydy@alibaba.com	Female	Oued
Lill					

	country	purpose_category	Risk Level	FICO Category
0	Portugal	Financial	Low Risk	Excellent
1	Portugal	Financial	Low Risk	Excellent
2	Indonesia	Other	Low Risk	Very Good
3	Japan	Educational/Business	Low Risk	Very Good
5	Tunisia	Financial	Low Risk	Very Good

[5 rows x 26 columns]

Identify customers with more than average inquiries and derogatories records

```
def identify_high_inq_derog(row):
    average_inq = complete_data['inq.last.6mths'].mean()
    average_derog = complete_data['pub.rec'].mean()

    if row['inq.last.6mths'] > average_inq and row['pub.rec'] >
average_derog:
        return True
    else:
        return False
```

```
complete_data['High_Inquiries_and_Public_Records'] =
complete_data.apply(identify_high_inq_derog, axis=1)
```

```
complete_data.head()
```

	loan_id	customerid	credit.policy	purpose	int.rate	\
0	193	C193	1	debt_consolidation	0.0743	
1	114	C114	1	debt_consolidation	0.0712	
2	920	C920	1	all_other	0.0800	
3	56	C56	1	small_business	0.1059	
5	503	C503	1	credit_card	0.0832	

	installment	log.annual.inc	dti	fico	days.with.cr.line	...	\
0	178.68	11.461632	1.20	812	16259.04167	...	
1	64.96	11.608236	0.22	812	16213.00000	...	
2	313.37	10.274672	2.24	792	14100.00000	...	
3	398.69	11.002100	17.12	747	14008.95833	...	
5	160.57	10.985293	7.08	767	13319.04167	...	

first_name	last_name	email	gender
------------	-----------	-------	--------

city \					
0	Yuma	Malecky	ymalecky5c@constantcontact.com	Male	
Lavradio					
1	Steven	Thalmann	sthalmann35@bloglines.com	Male	Rio
Meão					
2	Yetta	Cupitt	ycupittpj@etsy.com	Female	
Muruni					
3	Mycah	Drohun	mdrohun1j@paypal.com	Male	
Tendō					
5	Dorena	Keely	dkeelydy@alibaba.com	Female	Oued
Lill					

	country	purpose_category	Risk Level	FICO Category	\
0	Portugal	Financial	Low Risk	Excellent	
1	Portugal	Financial	Low Risk	Excellent	
2	Indonesia	Other	Low Risk	Very Good	
3	Japan	Educational/Business	Low Risk	Very Good	
5	Tunisia	Financial	Low Risk	Very Good	

	High_Inquiries_and_Public_Records
0	False
1	False
2	False
3	False
5	False

[5 rows x 27 columns]

```
class DataAnalysis:
```

```

    def __init__(self, df, column_name):
        self.df = df
        self.column_name = column_name

    def calculate_mean(self):
        return self.df[self.column_name].mean()

    def calculate_median(self):
        return self.df[self.column_name].median()

```

```

analysis = DataAnalysis(complete_data, 'fico')
print(analysis.calculate_mean())
print(analysis.calculate_median())

```

```

713.3056680161943
707.0

```


Bar chart to show distribution of loans by purpose

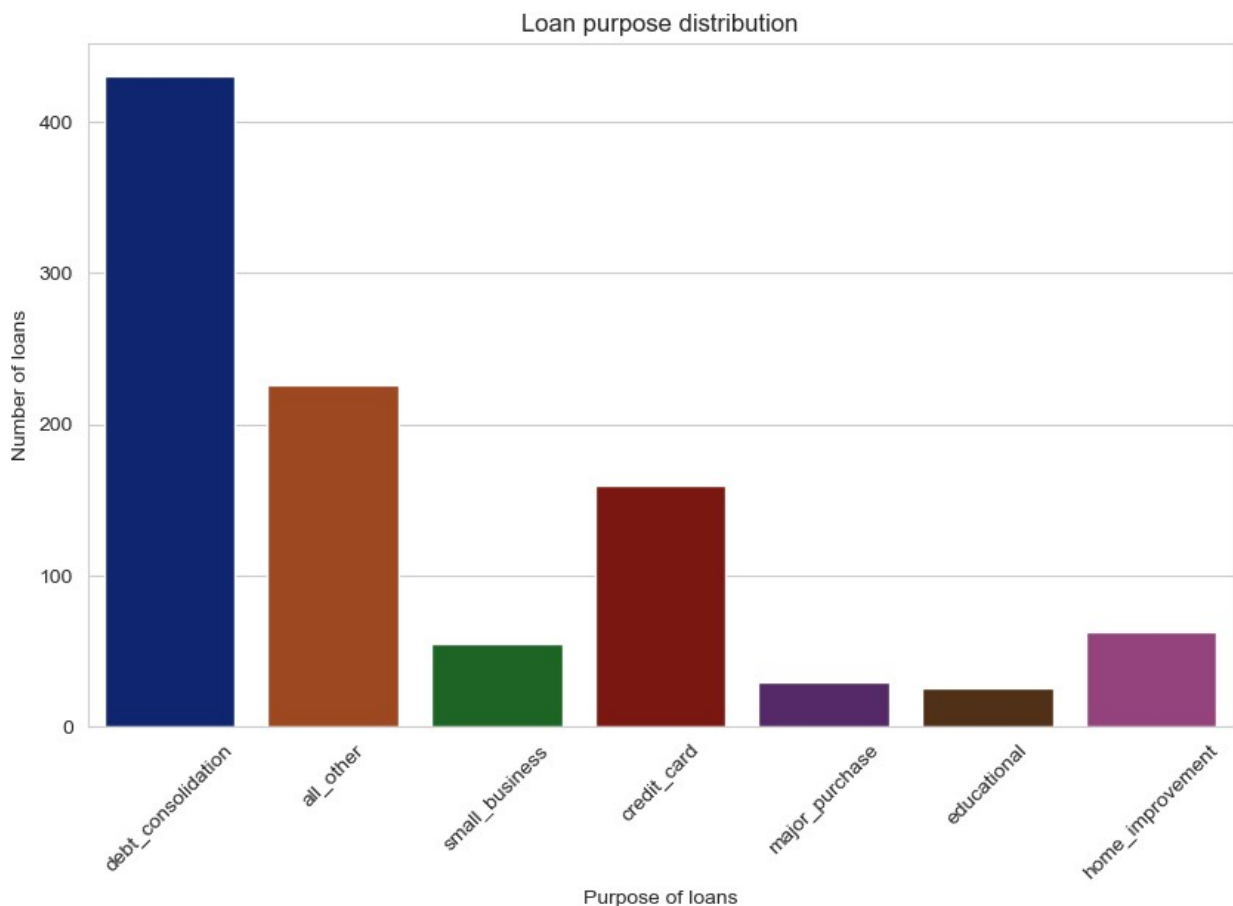
```
# Set the style of our visualization (darkgrid, whitegrid, dark, white)
sns.set_style('whitegrid')

plt.figure(figsize=(10, 6))
sns.countplot(x = 'purpose', data = complete_data, palette='dark')
plt.title('Loan purpose distribution')
plt.xlabel('Purpose of loans')
plt.ylabel('Number of loans')
plt.xticks(rotation=45)
plt.show()

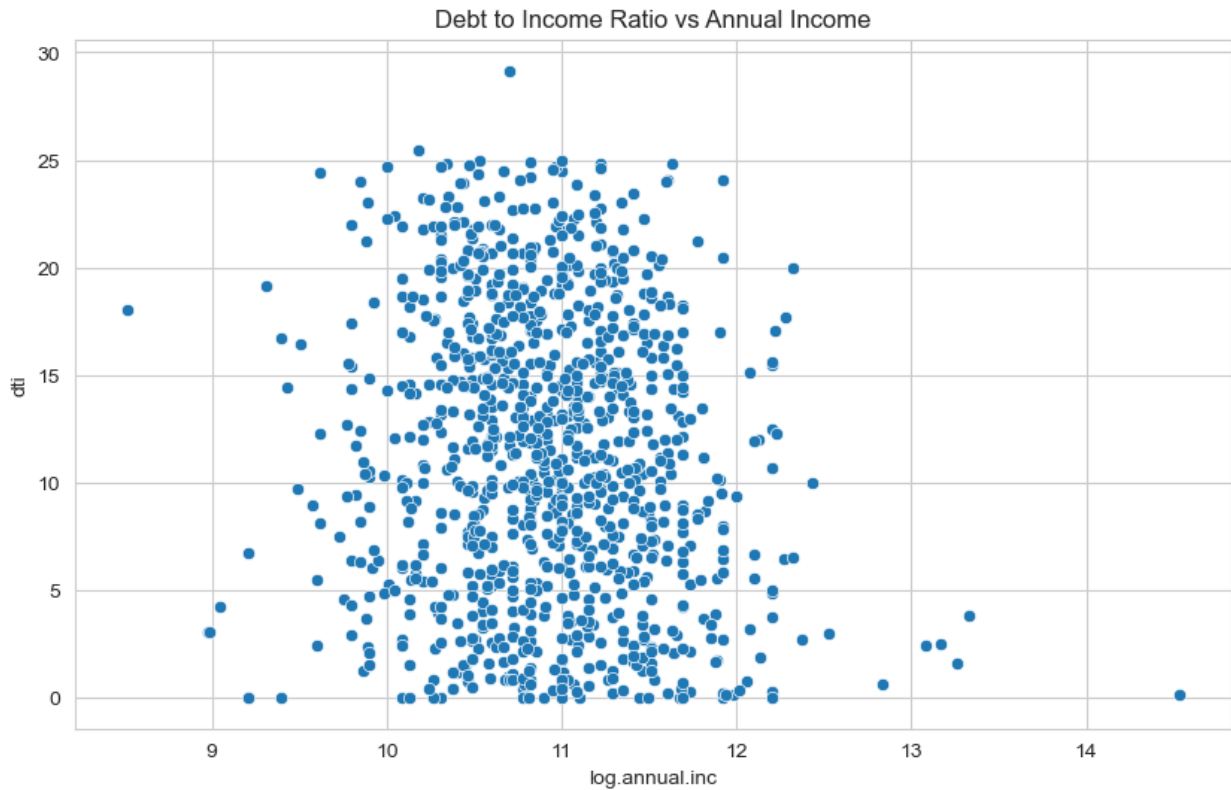
C:\Users\anasa\AppData\Local\Temp\ipykernel_4364\1856686829.py:5:
FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be
removed in v0.14.0. Assign the `x` variable to `hue` and set
`legend=False` for the same effect.

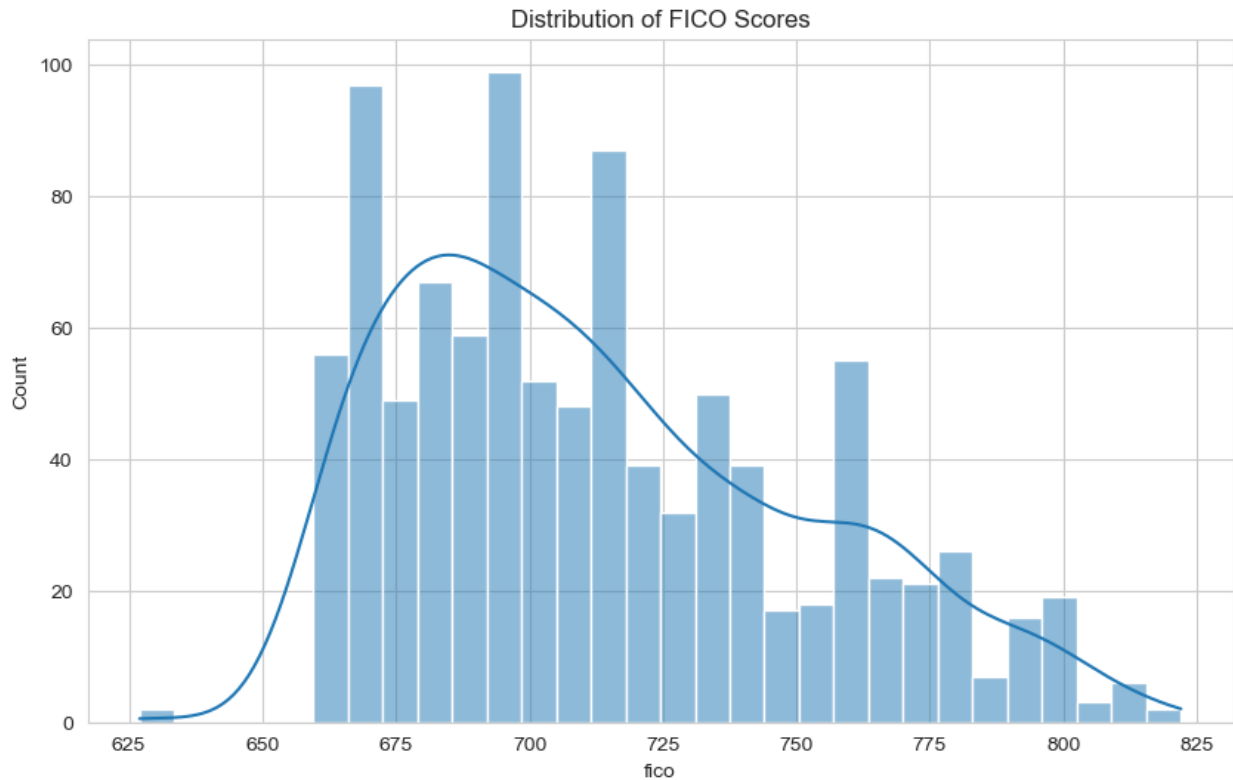
sns.countplot(x = 'purpose', data = complete_data, palette='dark')
```



```
# Create a scatterplot for 'dti' vs 'income'
plt.figure(figsize=(10, 6))
sns.scatterplot(x = 'log.annual.inc', y = 'dti', data = complete_data)
plt.title('Debt to Income Ratio vs Annual Income')
plt.show()
```



```
# Distribution of FICO score
plt.figure(figsize=(10, 6))
sns.histplot(complete_data['fico'], bins=30, kde=True)
plt.title('Distribution of FICO Scores')
plt.show()
```



Subplots

```
# Initialize the subplot figure
fig, axs = plt.subplots(2, 2, figsize=(20, 20))

# 1. Loan Purpose Distribution
sns.countplot(x='purpose', data=complete_data, ax=axs[0, 0])
axs[0, 0].set_title('Loan Purpose Distribution')
plt.setp(axs[0, 0].xaxis.get_majorticklabels(), rotation=45)

# 2. Debt-to-Income Ratio vs. FICO Score
sns.scatterplot(x='fico', y='dti', data=complete_data, ax=axs[0, 1])
axs[0, 1].set_title('Debt-to-Income Ratio vs. FICO Score')

# 3. Distribution of FICO Scores
sns.histplot(complete_data['fico'], bins=30, kde=True, ax=axs[1, 0])
axs[1, 0].set_title('Distribution of FICO Scores')

# 4. Risk Category vs Interest Rate
sns.boxplot(x='Risk Level', y='int.rate', data=complete_data,
ax=axs[1, 1])
axs[1, 1].set_title('Interest Rate vs. Risk Category')

# Adjust layout for readability
plt.tight_layout()
plt.show()
```

